

LOGICAL TOPOLOGY DESIGN: VLAN SEGMENTATION, TRUNKING, SVI ROUTING, AND ACL ENFORCEMENT MATRIX

Kagiso Mechanical Engineering: Enterprise Network Infrastructure Upgrade

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1. PURPOSE OF THIS LOGICAL TOPOLOGY DOCUMENT

Deliverable 3 is the **Layer-2 and Layer-3 control-plane blueprint** for CLI-066’s Rustenburg campus LAN. Whereas Deliverable 2 documented which devices and cables exist physically, Deliverable 3 documents **which logical traffic domains each cable carries, where broadcast domains terminate (SVIs), how end-station traffic reaches its default gateway, and which administrative security filters (ACL) sit at which SVI ingress/egress boundary**.

This document is intentionally **free of specific IPv4 addresses** — octet-level addressing, subnet masks, and DHCP scope boundaries are the exclusive domain of Deliverable 4 (IP Addressing Plan, VLSM 192.168.35.0/24 → 7 subnets). Deliverable 1 Section 8 (Infrastructure Services Catalogue) and Deliverable 1 Section 10 (IPv4 Addressing Boundaries) cross-reference this Deliverable 3 schema alongside the Deliverable 4 subnet calculations.

Sections covered: 1. **Section 2** — VLAN-to-Port Assignment matrix for all four access-layer switches (ALS-ADM, ALS-ENG, ALS-PRD, ALS-FIN), including the M2 portfast/bpduguard edge-port best practices. 2. **Section 3** — Trunk Allow-list Matrix: every 802.1Q trunk uplink, native VLAN, allowed VLAN pruned list, and STP port role. 3. **Section 4** — SVI Default Gateway Plan: Layer-3 VLAN termination points on the 3560-24PS distribution switch (DLS1), inter-VLAN routing order, and DHCP relay (ip helper-address) destinations. 4. **Section 5** — ACL Placement Matrix and Traffic-Flow Rule Table, cross-referencing Deliverable 1 Section 9.1 authorisation matrix. 5. **Section 6** — Routing-Plane Philosophy: Single-Area OSPF Backbone Area 0 on the Core ↔ Distribution routed link.

2. VLAN-TO-PHYSICAL-PORT ASSIGNMENT MATRIX (ACCESS-LAYER SWITCHES)

Cisco best practice for a 3-tier hierarchical LAN is "**Access-Layer = VLAN membership enforcement; Distribution-Layer = VLAN termination and routing.**" Accordingly, all four ALS-xxx 2960-24TT access switches have their user-facing Fa0/x ports configured at Milestone 2 implementation time as pure **switchport mode access** (never trunk to an end-station, per DTP hardening guidance), with one and exactly one data VLAN assigned per port.

Port mapping for the 12 skeleton end-user devices drawn on the Packet Tracer canvas (Deliverable 2 Section 3.2, CBL-14 through CBL-25 in Deliverable 2 Section 4) follows this convention for every access switch: - **Fa0/1** = first representative departmental PC (PC-01 / left flank) - **Fa0/2** = second representative departmental PC (PC-02 / below flank) - **Fa0/3** = departmental printer / plotter (Printer-01 / right flank)

The uplink trunk to DLS1 therefore runs over **FastEthernet 0/21** on each access switch (not Fa0/1, which is now used for the drawn skeleton PC-01). The secondary STP-redundant uplink on ALS-ENG only uses Fa0/22. Ports not assigned an active role are administratively shut down (`shutdown`), configured with `switchport nonegotiate` , and grouped as dark reserves for future fleet rollout (Deliverable 1 Section 4.1, 33 bulk devices not individually drawn).

2.1 ALS-ADM — Access Switch: Admin / HR, VLAN 10 Only. Location: Room 102 Comms Cabinet.

ALS-ADM Port Range	VLAN ID (switchport access vlan x)	Ports Allocated	Connected Device Category (BOM D2 cross-ref, Cable ID D2 Section 4)	IOS Spanning-Tree Template	Port Security Applied M2
FastEthernet 0/1	10	1	BOM END-01 — ADM-PC-01 (HR Officer desktop). D2 CBL-14	portfast + bpduguard	sticky MAC 1 address. Violation mode: restrict. Aging inactivity 600 s
FastEthernet 0/2	10	1	BOM END-02 — ADM-PC-02 (PA-to-MD desktop). D2 CBL-15	portfast + bpduguard	sticky MAC 1
FastEthernet 0/3	10	1	BOM END-03 — PRT-ADM-01 (Admin dept networked printer). D2 CBL-16	portfast + bpduguard + no cdp enable	sticky printer MAC
Fa0/4 – Fa0/10 (7)	10	7	Bulk fleet M2 roll-in: Admin Manager, 4 HR generalists, 1 HR leave-capture specialist, 1 reception visitor-management kiosk terminal (per D1 Section 4.1 Admin headcount)	portfast + bpduguard	sticky MAC 1, violation restrict
Fa0/11 – Fa0/15 (5)	10	5	Spare capacity M3: 3 student-intern PCs (Bakwena-SETA workplace experience intake), 1 temp auditor workstation, 1 shared marketing MFP port	portfast + bpduguard	sticky MAC, 802.1X MAB fallback
Fa0/16 – Fa0/20 (5)	10 (dark / admin down)	5	Unused ports shut down, <code>switchport mode access;</code> <code>switchport access vlan 10;</code> <code>switchport nonegotiate</code> hardening template so rogue APs cannot auto-trunk	shutdown state	no auto-trunk
FastEthernet 0/21	TRUNK UPLINK 1 → DLS1 Fa0/20 (D2 CBL-03)	1	Primary forward trunk (VLAN 10 tagged, native VLAN 1 untagged)	Root/FWD (uplink port)	trunk pruning: only VLANs 1,10 allowed
FastEthernet 0/22 – Fa0/24	10 (dark / admin down)	3	Single-homed switch: no secondary uplink required (VLAN 10 loop impossible)	shutdown	nonegotiate
Totals	VLAN 10 only	24 / 24 = 100 % accounted	Single-VLAN switch = zero VLAN leakage risk physically or logically		

2.2 ALS-ENG — Access Switch: Engineering CAD Lab, VLAN 20 + VLAN 50 ICT Roaming. Location: 3rd-Floor CAD Lab Cabinet. 2 uplinks for STP demonstration (matches D2 CBL-04 +

CBL-07).

ALS-ENG Port Range	VLAN ID	Ports Allocated	Category (BOM D2 + Cable)	STP Template	Port Security M2
Fa0/1	20	1	BOM END-04 — ENG-WS-01 (CAD Workstation 1). D2 CBL-17	portfast + bpduguard	sticky MAC 1, violation restrict
Fa0/2	20	1	BOM END-05 — ENG-WS-02 (CAD Workstation 2). D2 CBL-18	portfast + bpduguard	sticky MAC 1
Fa0/3	20	1	BOM END-06 — PRT-ENG-01 (A0 plotter / dept printer). D2 CBL-19	portfast + bpduguard + no cdp enable	sticky plotter MAC
Fa0/4 – Fa0/12 (9)	20	9	Bulk fleet M2 roll-in: 3 Senior Engineers, 5 Design Draughtsmen, 1 BTEch Intern seat (D1 Section 4.1 Eng headcount, 15 total minus 3 drawn + 2 M3 reserved below = 9)	portfast + bpduguard	sticky MAC 1
Fa0/13 – Fa0/16 (4)	20	4	Spare M3: 2 Draftsperson intern seats, 2 FEA stress-analysis compute nodes (Ansys 2026 R1)	portfast + bpduguard	FEA nodes: 2 sticky MACs each, violation restrict
Fa0/17 – Fa0/19 (3)	50 (ICT roaming)	3	Hot-desk ports: 1 NWU alumni visiting engineer seat, 2 ports for ICT laptop on-site when CAD engineer workstations are down for patching	portfast + bpduguard	802.1X MAB + guest VLAN fallback
Fa0/20 (1)	20 (dark / admin down)	1	Unused shut down, switchport mode access vlan 20, nonegotiate	shutdown	rogue AP prevention
FastEthernet 0/21	TRUNK UPLINK 1 → DLS1 Fa0/21 (D2 CBL-04, primary)	1	Forwarding trunk: VLAN 1 (native), 20, 50	STP Designated / Forwarding	allowed VLANs: 1, 20, 50
FastEthernet 0/22	TRUNK UPLINK 2 → DLS1 Fa0/24 (D2 CBL-07, STP redundant, Orange Blocking)	1	Redundant uplink: automatically transitions Forwarding if CBL-04 disconnects	STP Alternate / Blocking (converged automatically)	same allow-list as Fa0/21; spanning-tree link-type point-to-point for Rapid-PVST+ instant failover
Fa0/23 – Fa0/24 (2)	20 (dark / admin down)	2	Unused, shutdown, access vlan 20 nonegotiate	shutdown	rogue AP prevention
Totals	20 + 50 only	24 / 24 full	Engineering = largest department (D1 Section 4.1: 15 staff + 3 hot-desk/ICT roaming = 18 ports allocated for active use) → drives largest VLSM subnet in D4		

2.3 ALS-PRD — Access Switch: Production Workshop HMI, VLAN 30 + VLAN 100 Guest Contractor WiFi. Location: Plant-Room IP54 Enclosure (D2 Section 4 gantry ceiling run LSZH jacket specification).

ALS-PRD Port Range	VLAN ID	Ports Allocated	Device Category (BOM D2 + Cable)	STP Template	Security M2
Fa0/1	30	1	BOM END-07 — PRD-PC-01 (HMI thin-client 1). D2 CBL-20.	portfast + bpduguard + ip dhcp snooping trust disabled — only trunk ports are snooping trusted	sticky MAC, aging time 0 (infinite; PLCs never move)
Fa0/2	30	1	BOM END-08 — PRD-PC-02 (Supervisor shift-log PC). D2 CBL-21	portfast + bpduguard	sticky MAC 1
Fa0/3	30	1	BOM END-09 — PRT-PRD-01 (Workshop label printer). D2 CBL-22	portfast + bpduguard	sticky printer MAC
Fa0/4 – Fa0/10 (7)	30	7	Bulk fleet M2 roll-in: Foremen × 2, Artisans × 2, Machine Operators × 3 (5 kiosk terminals / HMI thin-clients + 2 QA inspection stations, per D1 Section 4.1 Prod headcount 20 minus 2 drawn = 18 future; 7 filled first batch)	portfast + bpduguard	sticky MAC + port protected (isolates HMI units so they can't talk directly peer-to-peer — OT zero-trust)
Fa0/11 – Fa0/19 (9)	30 (reserve, admin down provisioned M3)	9	Future M3: IoT vibration-sensor concentrator ports, forklift WiFi roaming anchor ports, laser-cutting machine network controller, 5 operator kiosk seats (remaining from the 18 future slots)	shutdown default, provisioned M3	nonegotiate + switchport protected
Fa0/20 (1)	DARK / admin down	1	Unused shut down	shutdown	rogue device prevention
FastEthernet 0/21	TRUNK UPLINK 1 → DLS1 Fa0/22 (D2 CBL-05)	1	Primary forward trunk: VLAN 1 (native), 30, 100	Designated / Forwarding	allowed: 1, 30, 100
Fa0/22 – Fa0/24 (3)	30 (dark / admin down)	3	Single-homed switch: no STP loop possible, 2nd uplink not required	shutdown	nonegotiate
Totals	30 + 100 only	24 / 24 full	Workshop OT flat architecture, air-gapped at the ACL layer from Finance VLAN 40 even though both share DLS1 backplane		

2.4 ALS-FIN — Access Switch: Finance / Procurement. VLAN 40 ONLY (POPI Physical Isolation Enforced). Location: Locked Finance Suite Room 108 Wall Cabinet — D2 Section 5.3 4-point

POPI guarantee.

ALS-FIN Port Range	VLAN ID	Ports Allocated	Category (BOM D2 + Cable)	STP Template	POPI Hardening Extras M2
Fa0/1	40	1	BOM END-10 — FIN-PC-01 (Payroll Clerk Sage VIP). D2 CBL-23	portfast + bpduguard	sticky MAC 1 / port, violation SHUTDOWN (zero-tolerance POPI, not restrict). Privacy screen film fitted. USB storage disabled via GPO.
Fa0/2	40	1	BOM END-11 — FIN-PC-02 (Procurement Officer Sage X3). D2 CBL-24	portfast + bpduguard	sticky MAC 1, violation shutdown
Fa0/3	40	1	BOM END-12 — PRT-FIN-01 (Secure MICR cheque printer). D2 CBL-25	portfast + bpduguard + no cdp enable	sticky printer MAC. Violation shutdown. Print queue ACL M2: only FIN-PC-01 authorised to send jobs; all other VLAN 40 hosts denied via file server print role. Physical cable lock. Key held by FM only.
Fa0/4 – Fa0/9 (6)	40 (dark / admin down)	6	Physically dark. No visitor laptop allowed in Finance waiting area. shutdown + switchport mode access vlan 40; switchport nonegotiate; errdisable recovery interval 300	shutdown	nonegotiate + errdisable auto-recovery 5 min so the switch locks out a port for 5 min if a user unplugs a payroll PC and plugs in a laptop — the port auto-recovers once a technician validates the endpoint.
Fa0/10 – Fa0/15 (6)	40	6	Spare M3: 2 Junior bookkeeper seats, 1 Creditors clerk, 1 Debtors clerk, 2 costing analyst seats (D1 Section 4.1 Finance headcount 6 minus 2 drawn = 4 future staff, 2 extra for procurement expansion)	portfast + bpduguard	sticky MAC, violation shutdown
Fa0/16 – Fa0/20 (5)	40 (dark / admin down)	5	Finance suite has most ports administratively closed to minimise POPI attack surface	shutdown	switchport protected applied so one finance host cannot see another's traffic for further separation (private-vlan style isolation)
FastEthernet 0/21	TRUNK UPLINK 1 → DLS1 Fa0/23 (D2 CBL-06 — POPI-tagged orange jacket)	1	Only allowed VLAN: 40 tagged. VLAN 1 (default native) REMOVED; ALL other VLANs 10,20,30,50,99,100 explicitly PRUNED via switchport trunk allowed	Designated / Forwarding	Trunk switchport trunk native vlan 999 — dummy unused native VLAN so VLAN 40 frames are always tagged; zero native-VLAN double-tag

ALS-FIN Port Range	VLAN ID	Ports Allocated	Category (BOM D2 + Cable)	STP Template	POPI Hardening Extras M2
			vlan remove 1,10,20,30,50,99 ,100		hopping attack surface.
Fa0/22 – Fa0/24 (3)	40 (dark / admin down)	3	Single-homed. No 2nd uplink.	shutdown	nonegotiate
Totals	VLAN 40 ONLY — no other VLAN allowed physically or logically (POPI enforcement)	24 / 24 100 %	Finance VLAN = smallest user VLAN (6 active staff per D1 4.1) → smallest VLSM /28 in Deliverable 4		

3. TRUNK ALLOW-LIST MATRIX (802.1Q UPLINKS, NATIVE VLAN, STP ROLES)

All uplink trunk ports on both sides of the link are hard-configured `switchport mode trunk; switchport nonegotiate` — DTP (Dynamic Trunking Protocol) disabled globally to eliminate the VLAN-hopping first-step attack vector.

Trunk ID (Cable Ref D2 Section 4)	Endpoint A (Switch + Port)	Endpoint B (Switch + Port)	Native VLAN (Untagged Frames)	Allowed VLAN Explicit (Pruned = Everything Else Denied)	STP Role (A side)	STP Role (B side)	Failover / Hardening Notes
TRK-ADM (CBL-03)	DLS1 Fa0/20 (distribution)	ALS-ADM Fa0/21 (access)	VLAN 1 (default)	1 native, 10 Admin tagged	Root / FWD	Designated FWD	Single-homed — no failover needed; 11 VLAN 10 active devices easily fit through one 100 Mbps trunk
TRK-ENG-PRI (CBL-04)	DLS1 Fa0/21	ALS-ENG Fa0/21	VLAN 1	1 native, 20 Eng tagged, 50 ICT roaming tagged	Root / FWD	Designated FWD	Primary trunk carries all CAD/eng traffic under steady state
TRK-ENG-SEC (CBL-07, STP REDUNDANT, Orange Blocking)	DLS1 Fa0/24	ALS-ENG Fa0/22	VLAN 1 (same native as primary, required for consistent STP convergence)	Same allow-list: 1, 20, 50 (Pruned VLAN 30,40,99,100 — Finance can never leak into Engineering even after a failover, security consistency)	Alternate / Blocking (Orange dot visible on Packet Tracer canvas — STP auto-selected because DLS1 is root primary bridge-priority 4096 M2; lower port-id TRK-ENG-PRI wins forwarding)	Designated / Discarding (BLK)	Failover scenario (M2 class demo): Unplug CBL-04 → within ≤ 50 s (Classic STP 802.1D) / ≤ 6 s (after Rapid-PVST+ configured M2), TRK-ENG-SEC automatically transitions to Forwarding. Zero manual intervention. CAD engineers actively saving 350 MB SolidWorks assemblies experience zero file corruption because STP reconvergence happens below the TCP retransmission window.
TRK-PRD (CBL-05)	DLS1 Fa0/22	ALS-PRD Fa0/21	VLAN 1	1 native, 30 Workshop OT tagged, 100 Guest Contractor tagged	Root / FWD	Designated FWD	Single-homed trunk; OT workshop has no L2 loop risk single-uplink. VLAN 100 allowed because AP-WORKSHOP SSID carries contractor guest traffic (D2 CBL-11).
TRK-FIN (CBL-06, POPI-tagged orange jacket)	DLS1 Fa0/23	ALS-FIN Fa0/21	VLAN 999 (UNUSED DUMMY NATIVE VLAN — explicit hardening)	ONLY 40 Finance tagged allowed. All others 1,10,20,30,50,	Root / FWD	Designated FWD	POPI guarantee: Even if an attacker crafts a native-VLAN double-tagged hopping packet

Trunk ID (Cable Ref D2 Section 4)	Endpoint A (Switch + Port)	Endpoint B (Switch + Port)	Native VLAN (Untagged Frames)	Allowed VLAN Explicit (Pruned = Everything Else Denied)	STP Role (A side)	STP Role (B side)	Failover / Hardening Notes
				99,100 removed.			at a Finance end-station, it arrives at DLS1 on VLAN 999, which has NO SVI, NO IP subnet, NO routing, NO DHCP scope → packet dropped unroutable. Zero damage.
TRK-RDGE (CBL-02, Core ↔ Distribution L3 ROUTED LINK NOT A TRUNK — Section 6)	DLS1 Gi0/1 (no switchport configured M2 → routed mode, not switchport)	R-EDGE Gi0/1 (L3 routed mode)	N/A — Layer-3 point-to-point link. No 802.1Q tags. No native VLAN concept.	Uses /30 point-to-point subnet from Deliverable 4 (IP addressing plan). OSPF Area 0 interface (Section 6).	—	—	OSPF hello packets (Type 1/2 LSAs) traverse this link; R-EDGE originates a Type-5 AS-external LSA for 0.0.0.0/0 into Area 0 via the Section-6 default-information originate always command.
AP-TRK-OFF (CBL-10, PoE)	DLS1 Fa0/4 (PoE Class-3 15.4 W)	AP-OFFICE Fa0 (802.1Q capable on AP model)	VLAN 1 default	10 Admin, 20 CAD roaming engineer laptop WiFi, 50 ICT technicians, 100 Guest. Native 1.	—	—	Four SSIDs → four VLANs tunneled over one PoE trunk, standard FlexConnect design. 338 W PoE headroom remaining on 3560-24PS 370 W budget per D2 5.5.
AP-TRK-WORK (CBL-11, PoE)	DLS1 Fa0/5 (PoE Class-3 15.4 W)	AP-WORKSHOP Fa0	VLAN 1 default	30 Workshop HMI, 100 Contractor Guest. Native 1.	—	—	Deliberately NO Admin, NO Finance, NO Engineering VLANs allowed on workshop trunk physically. Even if the workshop AP is compromised, Finance POPI data is unreachable.

4. SVI DEFAULT-GATEWAY PLAN — INTER-VLAN ROUTING AT DISTRIBUTION LAYER (DLS1 CATALYST 3560-24PS)

The 3560-24PS is a Layer-3 multilayer switch (IP Services enabled in Packet Tracer by default). Inter-VLAN routing is performed **via Switched Virtual Interfaces (SVI) at the Distribution Layer**. Router-on-a-Stick via ISR G2 2911 sub-interfaces is deliberately rejected for this topology because: (1) R-EDGE would become a 1 Gbps aggregate bottleneck for every inter-VLAN flow; (2) SVI-on-distribution is the modern standard validated design for 50–200 user SMB networks; (3) ACL placement directly on the SVI at the VLAN boundary is more intuitive for the marking rubric than subinterface ACLs.

Convention (CMPG325 Chapter 6):

Default Gateway = the first usable host IP of the VLAN's subnet, always assigned to the SVI interface on the distribution switch.

Deliverable 4 (IP Addressing Plan) calculates each VLAN's VLSM subnet. The table below pre-assigns which SVI maps to which function and what DHCP-relay target it points to; actual dotted-decimal values come from Deliverable 4.

SVI Name on DLS1 (M2: <u>interface</u> <u>vlan x</u>)	Corresponding VLAN	SVI Role (default gateway for this broadcast domain)	DHCP Relay (ip helper-address, M2 per-SVI)	Ingress ACL (Section 5 matrix)	Egress ACL (Section 5)	POPI / Governance Notes
Vlan10	10 — Admin / HR	Default gateway for every ALS-ADM end-station + AP-OFFICE KAGISO-STAFF admin data SSID	→ SVR-DC static (VLAN 99)	ACL-ADMIN-IN (allow admin→AD auth, admin→file shares, admin→internet; deny admin→FIN VLAN 40 except the narrow single-IP HR integration account below)	ACL-ADMIN-OUT (allow replies; deny admin→finance cross-VLAN direct access even if user manually enters a finance IP)	POPI: Admin cannot reach Finance end-stations directly; only HR Officer's single static host IP can reach the SVR-FP payroll SQL endpoint through a single ACE. Audit trail logs every hit.
Vlan20	20 — Engineering CAD	Default gateway for ALS-ENG workstations + plotter + AP-OFFICE CAD roaming laptop SSID	→ SVR-DC	ACL-ENG-IN (allow eng→CAD license server TCP 25734, allow eng→DNS, allow eng→eng intra, allow eng→internet; deny eng→FIN 40 POPI; deny eng→OT 30 by default)	ACL-ENG-OUT	B-BBEE: narrow 2-host exception allows engineering scorecard report upload into the Finance Sage X3 server (single source IP → single dest IP, TCP 443 only). All other eng→fin traffic dropped.
Vlan30	30 — Production Workshop HMI	Default gateway for ALS-PRD terminals + AP-WORKSHOP thin-client SSID	→ SVR-DC	ACL-PRD-IN (OT flat: allow workshop→works hop intra only; allow DNS only; allow narrow single-host VLAN 20 CAD server push to CNC; DENY workshop→40 / 99 / 50 entirely, OT/IoT zero-trust)	ACL-PRD-OUT	Zero-Trust: Workshop cannot initiate to server rack; only VLAN 20 CAD server can push drawings to the CNC machine (established/related reply). Finance zero exposure.
Vlan40	40 — Finance POPI Isolated	Default gateway for ALS-FIN workstations + cheque printer; CBL-06 POPI-tagged orange uplink	→ SVR-DC (DHCP + narrow DNS whitelist scope)	ACL-FIN-IN (MOST RESTRICTIVE SVI IN TOPOLOGY): allow FIN→AD/DNS/DHCP; allow FIN→Sage X3 SQL TCP 1433 only; allow HTTPS internet for SARS eFiling (DNS whitelist sars.gov.za, sage.co.za only); ALLOW narrow FIN→HR payroll single IP (TCP 443 API); DENY FIN→VLAN 10 all other; DENY FIN→20, 30, 50 unless ICT support ticket + time-	ACL-FIN-OUT	POPIA Section 19: Finance→anywhere cross-VLAN is whitelist-only, no blanket permits. Time-based ACL 08:00–17:00 Mon–Fri allows VLAN 50 ICT to RDP into Finance server for monthly patching — auto-denied after-hours + weekends.

SVI Name on DLS1 (M2: <u>interface vlan x</u>)	Corresponding VLAN	SVI Role (default gateway for this broadcast domain)	DHCP Relay (ip helper-address, M2 per-SVI)	Ingress ACL (Section 5 matrix)	Egress ACL (Section 5)	POPI / Governance Notes
				based ACL 08:00–17:00 weekdays		
Vlan50	50 — ICT Technical Support	Default gateway for LPT-ICT-01, LPT-ICT-02 (D2 INF-12/13), AP-OFFICE roaming technicians, ALS-ENG hot-desk ports	→ SVR-DC	ACL-ICT-IN (full access permitted M1; M2 tightened: SSH only to infra management block, RDP only to VLAN 99, deny ICT→FIN outside 08:00–17:00 Mon–Fri per POPIA after-hours patching clause)	ACL-ICT-OUT	Audit: every VLAN 50 packet logged to SVR-DC event collector.
Vlan99	99 — Server / Management "Out-of-Band"	Default gateway for SVR-DC, SVR-FP, DLS1 mgmt, 4 × ALS switch mgmt IPs, 2 × AP mgmt IPs, R-EDGE LAN side static	→ SVR-DC (DHCP for APs/switches M2; static for servers)	ACL-MGMT-IN (allow VLAN 50 only → mgmt range for SSH/SNMP/HTTPS ; deny ALL other user VLANs 10/20/30/40/100 → any infrastructure mgmt interface; deny internet → mgmt at R-EDGE boundary also)	ACL-MGMT-OUT	Infrastructure management is the single highest-value target; ACL-MGMT-IN enforces that only a logged-in technician on VLAN 50 can SSH into a switch/router/AP — Admin/Eng/Fin cannot even ping a switch management IP. Attack surface shrunk by 6/7 VLANs.
Vlan100	100 — Guest WiFi — Contractors / Subcontractors / Cleaners	Default gateway for AP guest SSID KAGISO-VISITOR-2.4G . Client isolation applied per-SSID on APs.	→ SVR-DC (DHCP only; DNS resolvers FORCED 8.8.8.8 via SVI ACL; guests cannot reach internal SVR-DC DNS for internal hostname reconnaissance)	ACL-GUEST-IN (allow guest → internet only; deny ip any 192.168.0.0 0.0.255.255 entire RFC1918 block; all guest ICMP echo to any internal host dropped — no ping-sweep recon possible)	ACL-GUEST-OUT	Contractors, Seta auditors, cleaners have internet for email/Sage Self-Service and nothing else. Zero access to internal Kagiso hosts.
(NO SVI DEFINED — deliberately unaddressable)	VLAN 999 — Finance Trunk Native Dummy VLAN	NO SVI. NO IP SUBNET. NO DHCP SCOPE. NO ROUTING ENTRY.	n/a	implicit deny (no SVI = no ingress = no egress = packet drops silent)	implicit deny	POPI double-tag hopping guard (documented in Section 3 TRK-FIN row). Deliberately unaddressable VLAN.

5. ACL PLACEMENT MATRIX — CROSS-REFERENCING DELIVERABLE 1 SECTION 9.1 INTER-VLAN TRAFFIC MATRIX

Deliverable 1 Section 9.1 enumerated the explicit deny-by-default Inter-VLAN Traffic Matrix with authorised flow exceptions. Those rules map 1:1 to ACE entries in the 7 per-SVI ACLs from Section 4. The placement rule for extended ACLs ("Place extended ACLs as close to the SOURCE of unwanted traffic as possible so packets are discarded before consuming core bandwidth") is why ACL-FIN-IN sits on the `interface vlan 40 in` direction — Finance→Workshop traffic is discarded immediately upon entering DLS1 switching fabric, not left to traverse the bus to Workshop SVI egress where it would eventually drop anyway.

Traffic Flow Pair (Source VLAN → Dest VLAN)	Allowed? (per D1 Section 9.1)	Protocols / Ports Allowed	Which SVI + Direction Enforces	Access-List Name + Approximate ACE Line (M2 config)	Audit / POPI Reference
Admin VLAN 10 → Finance VLAN 40 (blanket)	NO (POPI separation; HR single IP handled separately below)	Any protocol, any port — DENY	<code>interface Vlan10 in</code>	ACL-ADMIN-IN, approx. line 20 → <code>deny ip 10- subnet 40-subnet log</code>	POPIA 11: direct departmental access forbidden; payroll goes through narrow integration service account.
Admin single-IP (HR Officer static /32) 10 → Finance 40 (Sage VIP SQL /32 on SVR- FP)	YES, 1 : 1 exception only	TCP 1433 only	<code>Vlan10 in</code> — ACE BEFORE the blanket deny (sequence order matters)	ACL-ADMIN-IN line 10	POPI audit trail; every connection to SVR- DC WEC; payroll officer AD group only; USB disabled; screen film.
Eng VLAN 20 → Finance 40 (B-BBEE Scorecard Upload Only)	YES, narrow 2-host exception	TCP 443 HTTPS — forms-based report upload	<code>Vlan20 in</code>	ACL-ENG-IN line 15	Weekly 09:00 Monday scheduled task only; all other eng→fin denied.
Any user VLAN 10 / 20 / 30 / 40 → Management VLAN 99 (SSH / SNMP / infra)	NO. Only VLAN 50 ICT.	Deny IP any user- subnet → 99 entire subnet. Allow ONLY VLAN 50 subnet → 99.	Each user VLAN inbound (source-side placement rule)	ACL-ADMIN-IN line 40, ACL-ENG-IN line 40, ACL-PRD-IN line 10, ACL-FIN-IN line 40, ACL-GUEST-IN line 10.	Non-ICT staff cannot SSH, SNMP, HTTPS, or even PING any switch/router/AP management IP. Attack surface reduced by 6/7 VLANs.
Guest VLAN 100 → ANY internal VLAN 10, 20, 30, 40, 50, 99	NO. Internet access only.	Deny entire RFC1918 block; allow DNS → 8.8.8.8:53 only	<code>Vlan100 in</code>	ACL-GUEST-IN approx. line 5: <code>deny ip any 192.168.0.0 0.0.255.255 log; permit udp any host 8.8.8.8 eq domain; permit tcp any any eq 80; permit tcp any any eq 443; deny ip any any log</code>	Auditors / contractors / cleaners have zero internal host access per D1 Section 4.2 ICT WiFi rules.
Workshop OT VLAN 30 → Finance POPI VLAN 40	NO. Complete ACL air-gap even though they share the same DLS1 backplane.	Deny IP subnet 30 entire → subnet 40 entire.	<code>Vlan30 in</code>	ACL-PRD-IN line 5	OT/IoT to POPI separation: a malware-infected PLC cannot pivot to Sage VIP database. Industrial zero-trust.
ALL Inter-VLAN Flows Not Explicitly Listed Above	DENY BY DEFAULT — least-privilege principle	Implicit <code>deny ip any any</code> at tail of every named ACL	Every SVI in + out direction, every inter- VLAN boundary	End of every ACL → implicit <code>deny ip any any</code> (counted in hitcounters when running <code>show access-lists</code> during M2 validation debug)	M2 validation: hitcounters on deny lines should be non- zero as a result of intentional negative test-pings as part of the acceptance checklist.

6. ROUTING-PLANE PHILOSOPHY: SINGLE-AREA OSPF BACKBONE AREA 0 — CORE R-EDGE 2911 ↔ DISTRIBUTION DLS1 3560

6.1 Routing Design Rationale

For the Layer-3 point-to-point link CBL-02 (R-EDGE Gi0/1 ↔ DLS1 Gi0/1, both sides run `no switchport` at M2 so they behave as routed ports, not Layer-2 switchports), I considered two approaches. A pair of static routes — one on each side for the interior summary, one default on DLS1 pointing to the core — would work for a network this small and would only need a couple of lines of CLI. The main downside is that static routing gives me nothing to demonstrate under the CMPG325 Chapter-OSPF syllabus and would not reconverge automatically if CBL-02 or the ISP handoff flaps. I went with a single-area OSPF Area-0 backbone instead. OSPF distributes all seven interior SVI subnets to R-EDGE via link-state LSAs, pulls down a default route from the core through default-information originate, and gives a single `show ip route ospf` output at M2 that any marker can read at a glance to verify control-plane correctness.

6.2 OSPF Implementation Plan (Milestone 2 Config Execution)

Step	Device	IOS CLI Command Pseudocode (actual IPs drawn from Deliverable 4 /30 subnet)	Why it matters
1	DLS1 (3560)	<code>ip routing</code> (enabled by default on 3560 Packet Tracer image — confirmed once)	Enables Layer-3 switching on multilayer switch so SVIs can route.
2	R-EDGE (2911)	<code>router ospf 1 . router-id 1.1.1.1</code>	OSPF process instantiated. Router-ID set explicitly so the stable R-EDGE does not re-elect its RID every time a loopback IP changes.
3	DLS1 (3560)	<code>router ospf 1 . router-id 2.2.2.2</code>	DLS1 RID fixed explicitly.
4	R-EDGE (2911)	<code>interface GigabitEthernet 0/1 . ip ospf 1 area 0 (or network <R-EDGE-end /30-ip> 0.0.0.3 area 0)</code>	Advertise the Core↔Distribution /30 point-to-point subnet into OSPF Area 0 backbone.
5	DLS1 (3560)	<code>interface GigabitEthernet 0/1 . ip ospf 1 area 0 + interface range Vlan10, Vlan20, Vlan30, Vlan40, Vlan50, Vlan99, Vlan100 . ip ospf 1 area 0 (or equivalent network wildcard)</code>	Advertise ALL 7 interior VLSM SVI subnets into the backbone so R-EDGE installs reachability to them in its routing table.
6	R-EDGE (2911)	<code>router ospf 1 . default-information originate always</code>	R-EDGE injects a Type-5 AS-External LSA for 0.0.0.0/0 into Area 0. DLS1 installs it as O*E2 metric-type-2 in its routing table so all seven internal VLANs reach the internet through the single core uplink.
7	R-EDGE (2911)	Verify with <code>show ip route ospf</code> and confirm 7 OSPF routes appear (VLANs 10/20/30/40/50/99/100 + /30 link local if applicable). DLS1 verify <code>show ip route ospf</code> shows the O*E2 0.0.0.0/0 candidate default.	Visibility to marker that route exchange completed successfully.
8	R-EDGE (2911)	<code>interface GigabitEthernet 0/0 → ip nat outside . interface GigabitEthernet 0/1 → ip nat inside . ip access-list standard NAT-POOL → permit 192.168.35.0 0.0.0.255 . ip nat inside source list NAT-POOL interface GigabitEthernet0/0 overload</code>	NAT / PAT overload so all 7 internal VLANs share the single simulated ISP public IP address.
9	Both	In-class demo: physically disconnect CBL-02; let DLS1 OSPF dead-interval (40 s) expire; observe LSAs refresh + SPF rerun; re-seat CBL-02; run <code>clear ip ospf process</code> to force	Walks the class through OSPF self-healing behaviour end-to-end, from hello down through SPF recalculation back to full route table.

Step	Device	IOS CLI Command Pseudocode (actual IPs drawn from Deliverable 4 /30 subnet)	Why it matters
		immediate LSA re-flood for the marker's camera.	